Shared coefficient positions and integral Pell carriers

Two reductions for explicit Diophantine equations expressing Con(ZFC)

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AI-assisted mathematical research for the ProveIt repository

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Abstract

We investigate simpler fixed integer polynomials whose absence of zeros in their stated integer domains is provably equivalent in ZFC to syntactic consistency of ZFC. The comparison uses ProveIt commit `beb0192cb8dc36f78c369c080f2adfbc1887a486`, including the six wave-eight packages that are newer than the collection overview. Two complementary changes are developed. Compatible equal coefficient contributions may share an encoding position, reducing the degree and arithmetic cost of the nine-positive-variable construction, and also improving the ten-unrestricted-integer construction. A second reduction uses one integral Pell root to recover, in order, an index congruence and a missing scale divisibility; it gives a degree-26 equation in fourteen positive variables. All coefficients are fixed by explicit finite recipes. The archive contains literal arithmetic circuits, coefficient data, proofs, and independently written finite checks. The new reductions are ordinary mathematical proofs using named inherited compiler results; they do not constitute a new end-to-end proof-assistant certificate or a proof of global minimality.

Principal constructions

Witness domain	Unknowns	Exact degree	Binary operations
Strictly positive integers	9	1,675,396	458
Unrestricted integers	10	7,428,120	525
Strictly positive integers	14	26	92

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1 Results and the meaning of simplicity

The intended logical statement is

$$\mathsf{ZFC} \vdash \left(\neg\exists \mathbf{x} \in \mathbb{Z}^n_{>0}\ P(\mathbf{x}) = 0\right) \iff \mathrm{Con}(\mathsf{ZFC}). \tag{1}$$

Here Con(ZFC) means that there is no finite derivation of falsity from the fixed syntactic ZFC axiom predicate used by ProveIt. No consistency assumption enters the construction. In particular, proving the equivalence does not prove either side of it.

There is no agreed total order on polynomial complexity. We report the witness domain, number of witnesses, exact total degree, arithmetic circuit size, and coefficient binding separately. A circuit operation is one live binary addition, subtraction, or multiplication, including multiplication by a fixed numeral. Reusing a register costs nothing. Each fixed power is implemented by ordinary multiplication gates. Coefficient leaves are fixed integers, and their bit lengths are a separate cost.

Construction	Positive variables	Exact degree	Operations
Package 59, recommended nine-variable form [9]	9	1,910,404	473
Shared positions, primary form	9	1,675,396	458
Shared positions, cheaper tradeoff	9	1,738,228	457
Package 59, fourteen-variable form [9]	14	52	79
Integral carriers, primary form	14	26	92
Integral carriers, cheaper tradeoff	14	30	85
Package 58, degree-24 fifteen-variable form [8]	15	24	90
Package 51, 66-operation positive circuit [4]	15	157	66

Table 1: Comparisons at the pinned repository snapshot. Every row is a single polynomial equation over strictly positive integers. Earlier rows are cited constructions, not newly rerun end-to-end semantic proofs.

The primary nine-variable change lowers degree by approximately 12.30% and operation count by approximately 3.17% relative to package 59. The fourteen-variable degree falls by 50%, at the cost of thirteen additional gates. The degree-24 fifteen-variable equation and the 66-operation equation retain their distinct advantages. Neither new family reduces the smallest variable count in the inspected collection. These are concrete improvements to that collection, not a claim that the longstanding problem of the globally simplest consistency equation has been solved.

Over ten unrestricted integer variables, the shared-position adapter has exact degree 7,428,120 and 525 operations. It improves on package 59's 8,469,528 at 533 operations, and on the degree of package 57's 8,317,400 at 544 operations [7, 9]. The older package 54 point uses fewer operations (513), at degree 25,838,224 [5]; we therefore do not claim the least arithmetic cost in that domain. This third construction uses the integer-domain finalizer based on Sun's ten-unknown theorem [12], with its hypotheses and bootstrap inherited from the pinned package 59. In its version of (1), the quantified domain is $\mathbb{Z}^{10}$.

The constructions deliberately retain two different complete coefficient bindings. The nine-variable and ten-integer families use the original source-31 tuple; the fourteen-variable family uses the compact source-44 tuple. Coefficient ports with the same names must not be exchanged between them. Their bindings, hashes, and regeneration instructions accompany the circuits.

2 Logical scope and inherited interfaces

The existing ProveIt pipeline relates a literal finite computation to a syntactic ZFC refutation, translates that computation into arithmetic, and encodes the arithmetic using Pell equations or coefficient extraction. Our work replaces specific arithmetic reductions within that pipeline. The proofs below establish existence equivalences with the supplied source systems. Composing those equivalences with the cited native compiler and refutation calibration gives (1).

Three levels of evidence must be kept distinct. The mathematical proofs establish universal implications over integers. The finite programs check literal circuit transcription, all finite source identities, exact-degree certificates, coefficient hashes, and regeneration. A proof-assistant kernel certificate for a complete closed consistency equation is a separate artifact. ProveIt's active research records such closed constructions, but those theorems do not automatically certify the new literal polynomials here. We do not supply a newly checked object-language ZFC derivation or a complete new Lean/Rocq build.

All operations needed by the new proofs are elementary arithmetic, finite polynomial identities, the stated Pell results, and the explicitly identified inherited computation theorem. They are formalizable in ZFC. The imported files retain their own statements of scope and provenance. The final review records any remaining reliance on those interfaces.

3 Why shared coefficient positions are valid

A useful way to encode several equations is to extract chosen coefficients from a product of polynomials. Let

$$Z(B,\mathbf{x}) = \sum_{i=1}^{m} x_i B^{w_i}, \qquad A_d(B) = \sum_{u} a_{d,u} B^{u}.$$

At a chosen target t_j, the degree-d contribution is

$$[B^{t_j}]A_d(B)Z(B,\mathbf{x})^d = \sum_{\substack{u+\mathbf{w}\cdot\boldsymbol{\alpha}=t_j \\ |\boldsymbol{\alpha}|=d}} a_{d,u}\binom{d}{\boldsymbol{\alpha}}\mathbf{x}^{\boldsymbol{\alpha}}. \tag{2}$$

The coefficient ring can itself be a polynomial ring in named fixed coefficient ports. Thus equality of two coefficient contributions can be checked symbolically before enormous numerals are substituted.

Proposition 3.1 (Sharing a compatible coefficient position)**.** *Suppose the right side of (2) equals the required source polynomial for each target and each homogeneous degree. Then the coefficient decoder is correct, regardless of whether a kernel position serves more than one target. When several requirements at the same position ask for the identical coefficient, that coefficient is stored once.*

Proof. Expand the finite product and collect powers of B. Formula (2) enumerates every contribution, including all interactions with every monomial in Z^d. A kernel position contains one coefficient. Each target sees that coefficient whenever the exponent condition holds. Therefore the checked identities establish the decoder directly; no assumption that every target has private kernel positions is needed. Storing repeated requirements by addition would instead change the coefficient and is not the construction asserted here. □

This is a relaxation of the former placement constraint. It does not permit arbitrary collisions. The independent checker enumerates every relevant interaction and compares complete source

polynomials. The following section supplies the finite certificate, the complete polynomial definition, and the transport of the source bounds to the final equation.

4 Nine variables: compatible kernel coefficients

This contribution changes the source encoding in package 59 while retaining its arithmetic finalizer. It gives two explicit integer polynomials on nine strictly positive integer inputs:

Construction	Exact degree	Binary operations	Source height
Package 59, inherited comparator	1,910,404	473	2161
Shared coefficients, primary	1,675,396	458	1873
Shared coefficients, economy	1,738,228	457	1950

The primary lowers degree by approximately 12.30% and operation count by 3.17% against this comparator. The two new points are mutually nondominated. They retain the existing variable count, the entire original native coefficient tuple, and the same unbounded compiler and ratio-theorem dependencies. These results constitute a measured local improvement; they do not establish a global minimum.

4.1 The shared-coefficient lemma

Let $r_1, \dots, r_{22}$ be the original quadratic source residuals in 31 positive coordinates, specified by the literal 65-gate syntax in `nine/data/source_system65.json`. Put

$$\rho_{14} = r_{14} + r_2, \qquad \rho_{15} = r_{15} + r_3, \qquad \rho_j = r_j \quad (j \neq 14, 15).$$

These row operations are invertible over the integers. For a monomial p^α of ordinary degree $d \leq 2$ with coefficient $a_{j,\alpha}$ in ρ_j, define

$$c_{j,\alpha} = \begin{cases} 2a_{j,\alpha}, & d = 0, 1, \\ a_{j,\alpha} \prod_i \alpha_i!, & d = 2. \end{cases}$$

Coefficients of absent monomials are zero. Choose positive distinct weights w_i and distinct targets q_j.

Lemma 4.1 (Compatible kernel coefficients)**.** *Suppose that, for $d = 0, 1, 2$, a finitely supported table k_d on nonnegative integers satisfies*

$$k_d(q_j - \alpha \cdot w) = c_{j,\alpha} \tag{$*$}$$

for every degree-d monomial, with $k_d(e) = 0$ for $e < 0$. Define

$$D_d(T) = \sum_e k_d(e) T^e, \qquad U_p(T) = \sum_i p_i T^{w_i}.$$

Then

$$[T^{q_j}]\big(D_2(T)U_p(T)^2 + D_1(T)U_p(T) + D_0(T)\big) = 2\rho_j(p).$$

Repeated requirements in $(*)$ may share a position whenever their complete coefficient polynomials agree. The shared coefficient occurs once in D_d.

Proof. The coefficient of p^α in the target is $k_d(q_j - \alpha \cdot w)$ times the multinomial coefficient in U_p^d. For $d = 2$ the latter is 1 on a square and 2 on a cross product, giving $2a_{j,\alpha}$ with the stated weighting. Degrees zero and one are immediate. Checking absent monomials excludes every unwanted contribution. □

The earlier construction prohibited every same-degree overlap of source records. The lemma replaces this prohibition by exact compatibility. It is essential to store an agreed coefficient once: summing its repeated requirements changes the polynomial and invalidates the target identity.

4.2 Explicit finite data and source bounds

Both certificates use the unchanged weight vector, in source-variable order,

$$w = (1, 7, 61, 36, 16, 85, 20, 67, 43, 11, 34, 13, 96, 80, 29, 50,\\ 2, 109, 110, 112, 116, 91, 53, 55, 27, 101, 92, 9, 107, 37, 60).$$

The primary targets in row order $1, \dots, 22$ are

$$q = (1117, 28, 1442, 1045, 1841, 1330, 1819, 614, 267, 1384, 1379,\\ 889, 337, 1592, 413, 673, 1115, 772, 1107, 1413, 1031, 85).$$

Its 89 weighted requirements use 84 distinct kernel coefficients. The nontrivial groups are as follows; coordinate names are the original source names, not the nine final inputs.

Degree	Position	Value	Shared requirements
1	1324	2	Row 11: $C2$; row 10: $Delta$
2	39	-1	Row 22: $g\,Baux$; row 9: ca
2	588	2	Row 8: τ^2; row 18: V^2
2	1032	-1	Row 4: sN; row 19: $i\,C2$; row 17: fo

The complete kernel tables are in `nine/data/kernels.json`. The economy tables are separate files. The checker derives the residuals independently from the original source syntax and verifies all $22(1 + 31 + 496) = 11{,}616$ coefficient requirements for each certificate, including absent monomials. Thus this finite certificate verifies universal polynomial identities rather than agreement at sampled assignments.

Write $W = \max w_i$, $Q = \max q_j$, and $H = \max_d(\deg D_d + dW)$. The exact values are

Variant	W	Q	H	$(\deg D_0, \deg D_1, \deg D_2)$	Unique taps
Primary	116	1841	1873	$(1384, 1739, 1641)$	84
Economy	116	1918	1950	$(1508, 1816, 1718)$	86

Both quadratic kernels have leading coefficient $+1$.

Use the complete original31 tuple, whose port bit bounds are

$$(1064921, 1070004, 1064921, 1064922).$$

Its compressed payloads, builder, transition input and inherited notices accompany the article. Every weighted kernel coefficient has absolute value below $M_c = 2^{2134927}$. Define

$$\kappa = 2134939, \qquad a = 2^\kappa = 4096M_c, \qquad B = ab^2.$$

For positive b, g, set

$$\ell = \sum_i B^{w_i}, \qquad U_s = \ell + g, \qquad G_s = (D_2(B)U_s + D_1(B))U_s + D_0(B),$$
$$T_* = \sum_j B^{q_j}, \qquad N_0 = B^{117}, \qquad Z = B^{H+1},$$
$$M_* = N_0 - 1 - (b-1)\ell, \quad S_* = g + N_0(2G_s + T_*), \quad T_c = M_* + N_0(B-2)T_*.$$

Also put $t = bB^{116}$ and $G_{\mathrm{cap}} = 32t$.

Here $U_s \geq B^{116}$, and $D_2(B) > \frac{1}{2}B^{\deg D_2}$. Relative to $B^{\deg D_2}U_s^2$, the possibly negative terms have absolute values less than $2M_c/B^{18}$ and, respectively, $2M_c/B^{489}$ or $2M_c/B^{442}$. Both are less than $1/8$. Hence $G_s > 0$ before digit recovery. On the ratio-derived range $g < 4bB^{116}$, one has $U_s < 6bB^{116}$ and

$$\frac{2G_s}{Z} < \frac{86M_cb^2}{B-1} < \frac{1}{32}.$$

Since $H - Q = 32$, both T_* and $(B-2)T_*$ are smaller than $Z/64$. Together with $g < N_0$, this gives $S_* + T_c < B^{H+118}$.

When b is a power of two and $S_* \,\&\, T_c = 0$, the low mask recovers $g = \sum_i z_i B^{w_i}$ with $0 \leq z_i < b$, and U_s encodes $p_i = z_i + 1 > 0$. Every convolution coefficient has absolute value below

$$M_c\big((\textstyle\sum_i p_i)^2 + \sum_i p_i + 1\big) \leq 993M_cb^2 < B/4.$$

The inherited sparse-borrow lemma therefore turns $S_* \,\&\, T_c = 0$ into exactly the common source zero condition. Compatible sharing does not alter this bound: it concerns unique kernel coefficients. Conversely, every source zero works for sufficiently large admissible bases; the all-one source tuple fails row 5, so $g > 0$, and $g < bB^{116}$ gives $32g < G_{\mathrm{cap}}$.

4.3 Binary calibration and the complete final equation

Let $E_c = H + 118$ and $A_c = H + 255$. Define the fixed integers

$$\delta = (A_c - 15) \bmod 2, \qquad \lambda = (22 - A_c\kappa) \bmod 2,$$
$$n_B = (A_c - 15 - \delta)/2, \qquad n_2 = ((15+\delta)\kappa + \lambda - 22)/2.$$

Then

$$L = 2^\lambda bB^{E_c}, \qquad R = L(S_* + T_c + 1) - S_* - 1, \qquad N = 2^{n_2}b^\delta B^{n_B}.$$

The primary parameters $(E_c, \lambda, \delta, n_B, n_2)$ are $(1991, 0, 1, 1057, 17079501)$; the economy parameters are $(2068, 1, 0, 1096, 16012032)$. If $b = 2^k$, $B = 2^v$, $L = 2^u$, the two-block population identity is

$$\mathrm{pop}(R) = u + \mathrm{pop}(T_c) - \tau, \qquad \mathrm{pop}(T_c) = 139v - 31k - 22,$$

where $\tau \geq 0$ vanishes exactly for disjoint source words. The parameters give $u + \mathrm{pop}(T_c) = 2\log_2 N$. Legendre's valuation formula therefore gives $N^2 \mid \binom{2R}{R}$ exactly when the source words are disjoint. Both parity corrections are part of the definition.

The final inputs are $(b, g, h, s, w, C, i, j, n) \in \mathbb{Z}^9_{>0}$; put $j_0 = j - 1$. The complete remaining

registers are

$$\begin{aligned}
U &= N^2 w, & Y &= N^2 s,\\
M_0 &= RY, & P &= 2M_0^2 U,\\
K &= R + 1 + h(P-1), & A &= M_0(U+1),\\
B_1 &= 2R+1, & W_1 &= bw,\\
J &= 4A - 5, & Q_1 &= 3W_1 C - 2(W_1^2 - 1),\\
D &= (A^2-1)C^2 + 1, & k_0 &= iJC,\\
E_0' &= k_0 DC, & F &= (A^2-1)(E_0')^2 + 1,\\
G &= A + F(CD + 1 - A), & H_0 &= C + F(B_1 + j_0 C),\\
I &= (G^2-1)H_0^2 + 1, & &\\
a_0 &= (P^2-1)K^2 + 1, & a_1 &= DFI,
\end{aligned}$$

$$m_0 = K^2(4t - g) - 16t(C - KY)^2.$$

$$\begin{aligned}
V_0 &= PK, & V_1 &= k_0(8D^2 - 4D - 1)GH_0,\\
\Omega &= Jm_0, & E_0 &= J(n + W_1^2) + Q_1,\\
\Theta &= E_0 - \Omega(V_0 + V_1). & &
\end{aligned}$$

The desired single polynomial is

$$\boxed{\mathcal{F}_9 = [\Theta^2 - \Omega^2(a_0 + 64a_1)]^2 - 256\Omega^4 a_0 a_1.} \tag{3}$$

Every power has a fixed integer exponent, and every register is an abbreviation. The literal evaluator and every nonliteral fixed leaf are supplied. Each variant uses its own twelve-leaf binding; the native tuple is shared whole.

Proposition 4.2. *For each of the two fixed parameter sets, equation (3) has a positive integer zero exactly when the original fixed native source has a positive common zero. Composing with the inherited source compiler and calibration gives the stated ZFC-provable consistency equivalence.*

Proof. The compatible-coefficient lemma, the positivity and carry bounds, and the exact population calculation prove the same source contract as package 59. The arithmetic finalizer above is unchanged. Its proof, retained in `nine/inherited/nine_free_proof.md`, uses only that contract and the numerical conditions $R, N \geq 8$, $b \leq R, N$, which hold here. Its strict majorants first force $m_0 > 0$ from a norm zero. That recovers the carrier range before radical separation is invoked. Both radicals are then integral, $J \mid Q_1$ follows, and the Jones ratio theorem yields the binary base and binomial divisibility. The source contract recovers a common zero. Conversely, a source zero admits arbitrarily large admissible bases; the ratio theorem and exact CRT substitution supply the auxiliary witnesses, and its positive quotient construction supplies $n > 0$. No step uses separate ownership of a shared kernel coefficient or the old height 2161. □

4.4 Degree cancellation and reproducible evidence

Raw maximum-degree propagation through the norm is insufficient. Put $r = \deg R = 4E_c - 1$ and $q = \deg N = 2n_B + \delta$. With $T_p = k_0(8D^2 - 4D - 1)$, use the exact identities

$$\begin{aligned} T_p^2 - 64DF &= (8D+1)k_0^2 - 64D, \\ \Delta_1 := V_1^2 - 64a_1 &= \big((8D+1)k_0^2 - 64D\big)(GH_0)^2 + 64DF(H_0^2 - 1), \\ \Theta^2 - \Omega^2(a_0 + 64a_1) &= E_0^2 - 2E_0\Omega(V_0 + V_1) + \Omega^2\big(K^2 - 1 + \Delta_1 + 2V_0V_1\big). \end{aligned}$$

The unique leading term in the last line is $\Omega^2\Delta_1$, of degree $52r + 200q + 4W + 158$. The final subtracted term in (3) has degree $76r + 288q + 8W + 216$, strictly below twice that quantity. Therefore

$$\deg \mathcal{F}_9 = 104r + 400q + 8W + 316.$$

For $(r, q) = (7963, 2115)$ this is 1,675,396. The diagonal leading coefficient is $2^{186}a^{837288}c^{400} > 0$, where $c = 2^{17079501}$, proving that the upper bound is attained with the actual fixed tuple. For the economy parameters $(r, q) = (8271, 2192)$ the degree is 1,738,228.

The independent checker verifies the exact source identities, literal regeneration, actual coefficient payloads, operation recounts, the resolved leading forms and the strict degree comparison. It also compares the DAG with an independently written direct formula at modular assignments; these are transcription checks. The larger degree-certificate DAGs are not the evaluators counted as 458 or 457 gates. The translations to nonnegative coordinates require seven additional gates, giving 465 and 464.

All checks run with either ordinary Python or `python3 -O`; altered shared coefficients fail the mathematical identity checks. These checks are distinct from an end-to-end proof-assistant certificate and from an object-language ZFC derivation. The new finite source lemma is a natural next formalization target, followed by composition with the existing compiler endpoint. Joint optimization of row operations, coefficient equality classes and target positions is a separate further research question.

5 Two divisibilities and an index quotient in one Pell root

This section gives a second, direct route to smaller equations. It uses the authentic compact native tuple and ordinary Pell kernel, and does not use Jones's nine-variable compression. The new result has fourteen strictly positive unknowns, exact degree 26, and a 92-operation evaluator. A companion has fourteen positive unknowns, degree 30, and 85 operations. The inspected wave-eight fourteen-positive point is degree 52 at 79 operations. Thus the first construction halves its degree and adds thirteen gates; the second uses six additional gates and reduces its degree by 42.3%. These are tradeoffs within the pinned collection, with no claim of a global simplicity minimum.

The structural step combines the integral-root divisibility carrier of package 58 and the main-root index absorption of package 59. A single existing Pell root now enforces a weak congruence first, determines the Pell index, and only then enforces a stronger congruence restoring another divisibility. This ordered recovery permits the coordinate X to be supplied directly while deleting the old index quotient h.

5.1 Fixed coefficients and exact equation

Use the *whole* package-44 compact tuple, whose binding is included under `pell/coefficients/`. Its identifier is `wang943-complement67-shared-anchor-length8807-prime581197`. Set

$$B = 2^{590069}, \qquad \beta = B - 1, \qquad K_0 = D_C + BD_R, \qquad M_F = M_{F,\text{native}} + B - 1.$$

The fourth port is M_C. The high mask M_F is already shifted. The four little-endian binary integers occupy 295,388 raw bytes and 9,445 gzip bytes. Their recipes, payloads, exact hashes, literal machine data, native adapter, and independent coefficient auditor accompany the paper. All four ports are fixed numerals, not quantified coordinates.

The fourteen positive inputs are

$$j, F, \alpha, z, f, i, T, x, Y, r, \eta, \zeta, u, \gamma.$$

Define polynomial abbreviations

$$\begin{aligned} &J = 3j + 1, && q = \beta J + 1, && C = q - F - \alpha, && X = x + q, \\ &E = XY, && k = \eta + \zeta, && c = kY + \eta, && a = E + Y, \\ &H = 4a + 3, && \Delta = a^2 + H = (a+2)^2 - 1, && \ell = EkY. \end{aligned}$$

Also set

$$\begin{aligned} R &= [(q-1)(q-F) + \alpha](q^2 - 1) + (M_C + qM_F)J, \\ S &= 4qY, \qquad M = qS = 4q^2Y, \qquad v = k - R - 1, \\ D_* &= M(X + ac) + H(M\gamma - v + SX), \\ U &= \Delta(f^2 - 1), \qquad I = ic^2, \qquad V = f^2(cT - R) - c. \end{aligned}$$

The five residuals are

$$A_0 = [q^3(\ell + r) + Y]^2 - q^6[\ell(\ell + k) + 1], \tag{4}$$
$$A_m = D_*^2 - M^2(\Delta c^2 + 1), \tag{5}$$
$$A_t = q[K_0 C - F - z(q-1)] + XC, \tag{6}$$
$$A_a = (V + u)^2 - Uu(2V + u) - 1, \tag{7}$$
$$A_s = I^2 - U. \tag{8}$$

The single equation is

$$\boxed{P_{14,26} = A_0^2 + A_m^2 + A_t^2 + A_a^2 + A_s^2 = 0.} \tag{9}$$

There are no variable exponents, radicals, divisions or extra quantified registers in this definition. The supplied literal circuit is `pell/circuits/p14_degree26.json`.

Theorem 5.1 (Fourteen-positive equation)**.** *For the stated authentic coefficient tuple, $P_{14,26}$ has exact total degree 26 and a 92-gate evaluator. The new reductions below, composed with the explicitly supplied ordinary native compiler and calibration, establish by ordinary mathematics formalizable in ZFC that*

$$\mathsf{ZFC} \vdash [\neg\exists \mathbf{t} \in \mathbb{Z}_{>0}^{14}\ P_{14,26}(\mathbf{t}) = 0] \longleftrightarrow \mathrm{Con}(\mathsf{ZFC}).$$

This is not a newly kernel-checked end-to-end formalization or a supplied derivation file in the object-language ZFC calculus.

5.2 The two-stage carrier lemma

The elementary fact behind the construction is that a rational square root of an integer is integral: $m^2 \mid n^2$, with $m > 0$, implies $m \mid n$. At a zero of A_0, it gives

$$q^3 \mid Y, \qquad s = Y/q^3 > 0, \qquad \tau = \ell + r + s > 0, \qquad \tau^2 = \ell(\ell + k) + 1.$$

Since 590069 is odd, $\beta \equiv 1 \pmod 3$. Thus the supplied $J = 3j + 1$ gives $q \equiv 2 \pmod 3$. As $q \mid Y$,

$$H = 4Y(X + 1) + 3 \equiv 3 \pmod q, \qquad \gcd(q, H) = 1.$$

The odd integer H also satisfies $\gcd(Y, H) \mid 3$. Hence

$$\gcd(S, H) \mid 3, \qquad \mu := \frac{S}{\gcd(S, H)} \geq \frac{4qY}{3}. \tag{10}$$

For $Y \geq q^3$ and $R < q^4 - 2$, this yields $\mu > R + 1$. Moreover both S and M divide $4XY^2$; for M the quotient is $XY/q^2 = Xsq$, which is already integral without assuming $q \mid X$.

At a zero of A_m, the same rational-root lemma gives the integer $g = D_*/M$. Reducing $Mg = D_*$ only modulo S gives

$$S \mid H(k - R - 1), \qquad \mu \mid k - R - 1. \tag{11}$$

The Pell argument below uses this weaker congruence to recover the exact index and show $M \mid v$. Only then do we return to the full modulus:

$$M \mid HSX \quad \Longrightarrow \quad q \mid HX \quad \Longrightarrow \quad q \mid X. \tag{12}$$

The order of these steps prevents a circular use of the quotient X/q.

5.3 Soundness: recovering the indices before the quotients

At a zero of (9), all five residuals vanish. Restore s and τ as above. The cleared transport equation becomes the rational equality

$$(K_0 + X/q)C = F + z(q - 1).$$

Both its right side and its coefficient of C are positive. Therefore $C > 0$ before X/q has been shown integral. The authentic mask bounds give

$$\begin{gathered} q \geq 16, \qquad q(q^2 - 1) < R < q^4 - q^3 + q^2 - 1 < q^4 - 2, \\ R - 2 > 3q + 1, \qquad E > q^4 > R + 2, \qquad a > R + 2, \qquad a + 2 < 2E. \end{gathered} \tag{13}$$

For clarity, the bounds on R follow from

$$G = (q - 1)(q - F) + \alpha = q\alpha + (q - 1)C, \quad q \leq G \leq q^2 - q,$$

and $0 < (M_C + qM_F)J < (q - 1)(1 + 2q)$. The supplied shift $X = x + q > q$ is what proves $E > q^4$ at this stage. The inequalities $kY < c < k(Y + 1)$ follow directly from $\eta, \zeta > 0$.

The integer $g = D_*/M$ is positive. Indeed, $v < k$, $b_0 = X + ac > akY$ and $H < 5a$ give

$$D_* > Mb_0 - Hk > ak(MY - 5) > 0.$$

Hence $g^2 - \Delta c^2 = 1$. The first and main norms give positive Pell indices n, p with

$$\begin{aligned} &\tau = \chi_P(n), && k = 2\psi_P(n), && P = 2XY^2 + 1, \\ &g = \chi_A(p), && c = \psi_A(p), && A = a + 2. \end{aligned}$$

Here $\chi_A(t) + \psi_A(t)\sqrt{A^2-1} = (A+\sqrt{A^2-1})^t$. The first norm has fundamental integer-coordinate solution $(2XY^2+1, 2)$ for discriminant $XY^2(XY^2+1)$: a second coordinate 1 is excluded by consecutive squares. Since $P > A$ and $c > k$, monotonicity implies $p > n$.

The Pell recurrence gives $k \equiv 2n \pmod{4XY^2}$. Since $S \mid 4XY^2$, (11) implies

$$2n = R + 1 + t\mu \quad (t \in \mathbb{Z}).$$

The inequalities $0 < R+1 < \mu$ and $n > 0$ force $t \geq 0$. Thus $n > 24$ and $p \geq 26$. Consequently

$$c > A\Delta^2, \qquad c > 2p, \qquad c > 2R. \tag{14}$$

For the first estimate it suffices that $p \geq 6$ and $\psi_A(6) = 32A^5 - 32A^3 + 6A > A(A^2-1)^2$. The second is elementary Pell growth, and the third follows from $c \geq \psi_A(2) = 2A > 2R$.

The equation $A_s = 0$ gives $U = I^2$ with $I = ic^2 > 1$, hence $f \geq 2$. The *ordinary strong-rank lemma* supplied in `pell/upstream/ORDINARY_KERNEL.md`, Sections 4–5, now applies with all its hypotheses established. It gives an index m such that

$$f = \chi_A(m), \quad p \mid m, \quad c \mid m, \quad m > 2p, \quad I = \Delta\psi_A(m), \quad c^2 \mid f^2 - 1. \tag{15}$$

The last divisibility is a conclusion of the rank lemma; it cannot be obtained by informally cancelling Δ in $I^2 = \Delta(f^2-1)$. The copied proof handles the possibly nonfundamental squarefree Pell parameter and is an explicit inherited dependency.

The auxiliary equation is exactly

$$I^2V^2 - (I^2-1)(V+u)^2 = 1.$$

It excludes $V = 0$ and $V + u = 0$. Put $y = |V+u| > 0$. The supplied quotient and (15) imply

$$V \equiv -c \pmod{f}, \qquad V \equiv -R \pmod{c}.$$

Also $c^2 \mid I$. The signed auxiliary-index lemma in the same ordinary source gives an auxiliary index congruent to $\pm p$ and $\pm R$ modulo c. The bounds $0 < p, R < c/2$ force $p = R$. This argument does not assume $V > 0$. Its ordinary source quotient is restored by $T_{\text{old}} = fT > 0$.

Now $n < p = R$, so $|2n - R - 1| < R < \mu$. Together with the congruence modulo μ this yields $2n = R + 1$. Therefore

$$v = k - R - 1 = k - 2n$$

is a positive multiple of $4XY^2$. Positivity follows from $n \geq 2$ and $\psi_P(n) > n$. In particular $M \mid v$, and (12) proves $q \mid X$. Restore

$$w = X/q > 0, \qquad h = v/E > 0, \qquad \gamma_{\text{old}} = \gamma - v/M + w. \tag{16}$$

These are integers. The main root becomes $g = X + ac + H\gamma_{\text{old}}$. Since $n \geq 2$ gives $k \geq 4P > X$, one has $c > X$; and $g > \sqrt{\Delta}c > (a+1)c$. Thus $\gamma_{\text{old}} > (c - X)/H > 0$. The transport equation is now the genuine integer equality $(K_0 + w)C = F + z(q-1)$, corresponding to old source coordinate $z_{\text{old}} = z + 1$.

All positive ordinary source coordinates are recovered and all six source factors equal 1. The supplied native compiler and calibrated machine endpoint therefore yield a finite refutation of the specified ZFC calculus.

5.4 Completeness and the positive replacement witnesses

Conversely, a finite refutation gives canonical normalized witnesses from the authentic compact adapter. Transfer the strong coordinate by $i = \Delta i_{\mathrm{normalized}}$, as in package 58. Then $I^2 = \Delta(f^2-1)$, $c \mid I$, and $f^2 \equiv 1 \pmod{c}$. Canonical indices satisfy

$$p = R = 2n-1, \qquad n \geq 3, \qquad R \equiv 3 \pmod 4, \qquad X = 2^R, \qquad Y = sq^3,$$

and $k > XY$. These are inherited compiler/Pell facts, not conclusions of a finite numerical search.

The adapter supplies $q = B^N$ and $J = 1 + B + \cdots + B^{N-1}$ with odd $N = 581197^e > 1$. Since $B \equiv -1 \pmod 3$, $J \equiv 1 \pmod 3$ and $j = (J-1)/3 > 0$. Canonical $X = 2^R$ with $R > 3q$ gives $x = X - q > 0$. The source proves $w > q > F$, which makes $z_{\mathrm{old}} \geq 2$; hence the transport shift $z = z_{\mathrm{old}} - 1$ is positive.

Supply Y directly and set $r = \tau - \ell - s$. The first norm and $\ell > k$ give $\tau < \ell + k$ and

$$\tau - \ell = \frac{\ell k + 1}{\tau + \ell} > \frac{k}{3} > \frac{XY}{3} > s.$$

Thus $r > 0$ and $A_0 = 0$. The positive multiple $v = k - 2n$ of $4XY^2$ satisfies $M \mid v$. Set

$$\gamma = \gamma_{\mathrm{old}} + v/M - w.$$

This is positive for a particularly simple reason:

$$\frac{v}{M} \geq \frac{4XY^2}{4q^2Y} = \frac{wY}{q} \geq wq^2 > w.$$

The resulting D_* equals Mg, proving $A_m = 0$.

The auxiliary coordinates use the squared-modulus reconstruction already proved in package 58. For $R = 2r_0 + 1$ with r_0 odd, the elementary odd Pell polynomials satisfy

$$\chi_t(2r+1) = tQ_r(t^2), \qquad Q_r(0) = (-1)^r(2r+1),$$
$$Q_r(1-A^2) = (-1)^r \psi_A(2r+1).$$

Take $V_* = \chi_I(R)/I$ and $y_* = \psi_I(R)$. They are positive integers, $V_* > 1$, and $I^2V_*^2 - (I^2-1)y_*^2 = 1$. The identities yield $V_* \equiv -c \pmod{f^2}$ and $V_* \equiv -R \pmod c$. Since $\gcd(c, f) = 1$,

$$T = \frac{V_* + c + Rf^2}{cf^2}$$

is a positive integer and gives the required $V = V_*$. Finally $(I^2-1)(y_*^2 - V_*^2) = V_*^2 - 1 > 0$, so $u = y_* - V_* > 0$. All five residuals vanish. This proves both existence implications in Theorem 5.1 without changing the fixed tuple.

5.5 Exact degree and literal cost

The cancellation identities used in the degree proof are

$$A_0 = q^6[r^2 + \ell(2r-k) - 1] + 2q^3(\ell + r)Y + Y^2,$$
$$D_{\mathrm{old}} = X + ac + \gamma H, \qquad L = -v + SX,$$
$$A_m = M^2[(X + \gamma H)^2 + 2ac(X + \gamma H) - Hc^2 - 1]$$
$$+ 2MD_{\mathrm{old}}HL + H^2L^2.$$

These are polynomial-ring identities, not relations valid only on solutions. They give the exact residual degrees

$$(\deg A_0, \deg A_m, \deg A_t, \deg A_a, \deg A_s) = (11, 13, 3, 13, 10).$$

Put $q_0 = 3\beta j$, $X_0 = x + q_0$, $k_0 = \eta + \zeta$, and $R_4 = q_0^3(q_0 - F)$. The two highest residual forms are

$$\begin{aligned} L_m &= 32q_0^2 k_0 X_0^2 Y^4 (4q_0^2 Y\gamma + R_4), \\ L_a &= 2X_0^2 Y^2 f^4 u R_4. \end{aligned}$$

Hence the degree-26 homogeneous form is $L_m^2 + L_a^2$. The coefficient of $x^4Y^4f^8u^2j^6F^2$ is exactly $4(3\beta)^6 > 0$; L_m^2 contains no f or u and cannot change it. This establishes the exact degree for the authentic numerical instance.

The evaluator has 49 multiplications, 27 additions and 16 subtractions, totaling 92 live binary operations. Scalar multiplication is charged, every fixed power is expanded into multiplications, and all fourteen inputs are live. This arithmetic metric is separate from bit complexity and the height of the expanded coefficients.

5.6 A degree-30 companion with 85 operations

Retain the positive inputs

$$J, F, \alpha, z, f, i, T, w, Y, r, \eta, \zeta, u, \gamma,$$

and put $q = \beta J + 1$, $X = wq$. Use the same common expressions, but replace the main carrier and transport by

$$\begin{aligned} M &= 4qY, \qquad v = k - R - 1, \\ D_* &= M(X + ac) + H(M\gamma - v), \\ A_t &= (K_0 + w)C - F - z(q - 1). \end{aligned}$$

Keep A_0, A_m, A_a, A_s and the five-square sum, calling it $P_{14,30}$. The literal circuit is in `pell/circuits/p14_degree30.json`.

There is no restriction modulo 3 on J here. Once $q^3 \mid Y$, every prime common to qY and H is 3. If $3 \mid q$, then $27 \mid Y$ and $H \equiv 3 \pmod{27}$, so its 3-adic valuation is exactly one. If $3 \nmid q$, then $\gcd(q, H) = 1$ and $\gcd(Y, H) \mid 3$. As H is odd,

$$\gcd(4qY, H) \le 3, \qquad \frac{M}{\gcd(M, H)} \ge \frac{4qY}{3} > R + 1.$$

Also $M \mid 4XY^2$. The same ordered index proof gives $p = R$ and $2n = R + 1$, then $h = v/E > 0$ and $\gamma_{\text{old}} = \gamma - v/M > 0$. Completeness uses $\gamma = \gamma_{\text{old}} + v/M > 0$; both gap constructions remain unchanged. Thus it has the same consistency equivalence.

Its residual degrees are $(12, 14, 2, 15, 10)$. With $q_0 = \beta J$, $X_0 = wq_0$ and $R_4 = q_0^3(q_0 - F)$, the highest form is

$$[P_{14,30}]_{30} = 4X_0^4 Y^4 f^8 u^2 R_4^2.$$

The coefficient of $w^4J^{10}Y^4f^8u^2F^2$ is $4\beta^{10} > 0$. The evaluator has 45 multiplications, 24 additions and 16 subtractions, totaling 85 operations.

5.7 Verification and remaining formalization

The independent checker uses the standard Python library and does not import the circuit builders. It proves three local polynomial identities by exact sparse algebra, checks all gate dependencies and counts, and compares the DAGs with separately transcribed formulas at 200 modular assignments. It also checks 48 exact polynomial rays modulo two primes against the analytic leaders using the actual fixed tuple, exercises 10,584 bounded carrier cases, and rejects a transcription mutation for each circuit. The ordinary mathematical degree proof above supplies the universal upper bound and nonzero leading coefficient; the modular rays are implementation checks. The same checks pass under Python's `-O` mode.

The inherited coefficient audit independently rebuilds the four numerals, verifies 66,062 finite basis interactions and their provenance, tests 38,663 one-bit corruptions, and verifies 87,380 small complement models. These checks do not replace the unbounded native compiler proof. An independent internal mathematical review of the new carrier is included as `pell/INDEPENDENT_PROOF_AUDIT.md`; it is AI-assisted review, not external human peer review.

The next formalization target is the two-stage carrier lemma with its exact order of recovery: first the modulus S, then the index identity $p = R$, then the full modulus M and $q \mid X$. Potential further reductions must preserve this order and justify all positive replacement witnesses. Whether an auxiliary root can remove another coordinate without losing the degree advantage remains open.

6 A companion over ten unrestricted integers

The shared-kernel-tap source also composes with the unchanged Sun finalizer of package 59. This gives a fixed polynomial

$$F_{10}(f, g, h, k, l, w, x, y, z, m) \in \mathbb{Z}[f, g, h, k, l, w, x, y, z, m]$$

of **exact total degree** 7 428 120, with a division-free evaluator of **525 binary arithmetic operations**, such that the inherited compiler and arithmetic theorems, together with the new source argument, prove in ZFC

$$\neg\exists (f, g, h, k, l, w, x, y, z, m) \in \mathbb{Z}^{10}\ (F_{10} = 0) \quad \Longleftrightarrow \quad \mathrm{Con}(\mathrm{ZFC}).$$

The domain here is all integers. Relative to package 59's ten-integer polynomial of degree 8 469 528 with 533 gates, the degree decreases by 12.2959% and eight arithmetic operations are saved. The comparison is at commit `beb0192cb8dc36f78c369c080f2adfbc1887a486`.

6.1 The revised source polynomials

Use the primary shared-tap certificate, with its complete old31 coefficient tuple, the original unshifted 31-coordinate positive source, and the two invertible row operations. The 89 weighted source records use 84 unique kernel taps. Each shared coefficient is inserted once. The target identities are

$$[Z^{q_j}]\{D_2(Z)U_p(Z)^2 + D_1(Z)U_p(Z) + D_0(Z)\} = 2\rho_j(p), \qquad U_p(Z) = \sum_i p_i Z^{w_i}.$$

The certified numerical parameters are

$$W = 116, \quad Q = 1841, \quad H = 1873, \qquad (\deg D_0, \deg D_1, \deg D_2) = (1384, 1739, 1641).$$

The leading coefficient of D_2 is $+1$. Every kernel coefficient has absolute value below $C_0 = 2^{2134927}$. Put

$$\kappa = 2134940, \qquad \beta_S = 2^\kappa > 4096 C_0.$$

The even exponent is part of the source binding. Define

$$\begin{aligned} b &= 1 + 3f, & B &= \beta_S b^2, \\ \ell &= \sum_i B^{w_i}, & U_c &= \ell + g, \\ A_T &= \sum_j B^{q_j}, & N_0 &= B^{117}, \\ N_1 &= B^{1874}, & N &= N_0 N_1 = B^{1991}, \\ M_0 &= N_0 - 1 - (b-1)\ell, & J_c &= 2G_c + A_T, \\ S_c &= g + J_c N_0, & T_c &= M_0 + (B-2) A_T N_0, \\ \mathcal{C} &= bB^{116}. \end{aligned}$$

Here the remaining expressions are

$$\begin{aligned} G_c &= D_2(B) U_c^2 + D_1(B) U_c + D_0(B), \\ X &= N^2 + (S_c + T_c) N + N - 1 - S_c, \\ Y &= \beta_S^{15} 2^{\kappa - 21} B^{2114} b = 2^{2130\kappa - 21} b^{4229}. \end{aligned}$$

These are polynomial abbreviations, with fixed exponents. Their exact degrees are $7964, 4229, 233$ for $X, Y, \mathcal{C}$, respectively.

Suppose first that b is a square, $f \neq 0$, and $0 < g < 2\mathcal{C}$. Then $b \geq 4$, and $B^{116} \leq U_c < 3bB^{116}$. The kernel estimates give

$$D_2(B) > \tfrac{1}{2} B^{1641}, \qquad |D_1(B)| < 2C_0 B^{1739}, \qquad |D_0(B)| < 2C_0 B^{1384}.$$

After normalization by $B^{1641} U_c^2$, the two possibly negative terms are at most $2C_0/B^{18}$ and $2C_0/B^{489}$, each less than $1/8$. Consequently $G_c > \frac{1}{4} B^{1641} U_c^2 > 0$. The upper bound

$$|G_c| < 26 C_0 b^2 B^{1873}, \qquad 2G_c < N_1/64$$

and the 32-position gap $H - Q$ prove

$$0 < J_c < N_1/32, \qquad J_c + (B-2) A_T + 2 < N_1.$$

Since $0 \leq M_0 < N_0$ and $0 < g < N_0$, this yields the strict range

$$0 \leq S_c, \qquad 0 \leq T_c, \qquad S_c + T_c < N. \tag{17}$$

It also gives $X \geq N^2 > 3b$ and $Y > \max(b, 256)$. As κ is even, the source congruence is $X \equiv 1 - 22 \equiv 0 \pmod 3$.

When b is a power of two and $S_c \,\&\, T_c = 0$, the low mask recovers $g = \sum_i n_i B^{w_i}$ with $0 \leq n_i < b$. Thus U_c encodes positive coordinates $p_i = n_i + 1$. Every signed convolution coefficient h_s obeys

$$|h_s| < 993 C_0 b^2 < B/4.$$

The target padding in J_c and the mask $B - 2$ test $h_s = 0$ even when the incoming carry is -1. Therefore the same borrow argument proves

$$S_c \,\&\, T_c = 0 \quad \Longleftrightarrow \quad g \text{ encodes } n_i \in [0, b), \quad \rho_j(n+1) = 0 \text{ for all } j.$$

For $b = 2^k$ and $B = 2^v$, the actual mask population and block exponent are

$$s_2(T_c) = 139v - 31k - 22, \qquad \log_2 N = 1991v.$$

Hence

$$1 + \log_2 N + s_2(T_c) = 2130v - 31k - 21 = \log_2 Y.$$

Using (17), the exact two-block population identity and Legendre's formula turn $Y \mid \binom{2X}{X}$ into precisely the source zero condition. Conversely a positive source zero is not the all-one vector, since its fifth row has value one there. Choosing arbitrarily large $b = 4^s$ and encoding $n_i = p_i - 1$ gives $B \le g < \mathcal{C}$ and the required forward source witnesses.

6.2 The explicit unchanged finalizer

Here are the fixed polynomial formulas, with local notation distinguished from the source base and masks:

$$\begin{gathered}
L = lY, \quad U = 2LX, \quad V = 4gwY, \quad W_* = bw, \\
A = U(V+1), \quad B_* = 2X+1, \quad C = B_* + (A-2)h, \\
P = U^2V, \quad K = X + 1 + k(P-2), \\
S_* = 2A - 5, \quad T_* = 3W_*C - 2(W_*^2 - 1), \\
a_1 = (P^2-4)K^2 + 4, \\
\Delta_* = 8\mathcal{C}^3 gK^2 - g^2\{32(C-KL)^2\mathcal{C}^3 + g^2K^2\}, \\
O = f^2l^2\Delta_*, \quad \xi = zx, \quad a_0 = b\{1 + (3O-4)b^2z^2\}, \\
D = (A^2-4)C^2 + 4, \quad E = C^2D\xi, \quad F = 4(A^2-4)E^2 + 1, \\
G_* = 2(A+2)(A-2)^2E^2 - 1, \quad \Omega = 16E, \\
H_* = C + (B_* - C)F + \Omega Fy, \quad I_* = (G_*^2 - 1)H_*^2 + 1, \quad a_2 = DFI_*.
\end{gathered}$$

Finally set

$$\begin{aligned}
r &= T_* + S_*y, & \Sigma &= \xi S_*^5, \\
\Theta &= (r^3 - rS_*^2 - S_*^3)(r^2 + 23S_*^2), \\
w_0 &= 1, & w_1 &= a_0, & w_2 &= a_0(P^2 + K^2 + 3), \\
Z &= \Sigma m - \Theta, & A_i &= \Sigma^2 w_i^2 a_i \quad (0 \le i \le 2), \\
V_n &= Z^2 + A_2 - A_0 - A_1, & W_n &= V_n^2 + 4Z^2A_2 - 4A_0A_1.
\end{aligned}$$

The promised single polynomial is

$$\boxed{F_{10} = W_n^2 - 16V_n^2Z^2A_2.} \tag{18}$$

The inherited norm, separator, and intersective-quintic argument first gives $\Sigma \neq 0$, all three a_i squares, and $\Sigma \mid \Theta$. In particular $z \neq 0$. These conclusions use no source positivity. The coprime factors of the square a_0 must be signed squares. Since $b \equiv 1 \pmod 3$, it cannot be a negative square, so b is a positive square. The case $b = 1$ forces $f = O = 0$ and $a_0 = 1 - 4z^2 < 0$, impossible. Thus $b \ge 4$. The second factor being a square forces $O > 0$, since for $O \le 0$ it is at most $1 - 4b^2z^2 < 0$.

Now $O > 0$ implies $\Delta_* > 0$. As $\mathcal{C} > 0$, its displayed formula rules out $g \le 0$ and $K = 0$; division by $gK^2 > 0$ gives $g^3 < 8\mathcal{C}^3$. Hence $0 < g < 2\mathcal{C}$ before the source estimates are invoked. This

establishes the correct order of the signed bootstrap and explains why no additional positive padding term is needed.

Sun's inherited arithmetic bridge then recovers b a power of two and the binomial divisibility, giving a positive source zero. The converse uses the cofinal forward witnesses, the same Pell bridge, and residue interpolation. The precise dependencies are Sun's 2021 Theorems 4.12–4.13 [13] and their specialization in his 2026 Propositions 6.1–6.2 [12], as preserved in the package59 proof. Thus the new part is the verified source replacement; the compiler and finalizer theorem are inherited.

6.3 Exact degree certificate and finite checks

The upper bound is certified by induction at every gate of the literal division-free circuit: leaves have degrees zero or one, a product uses the sum of the two bounds, and a sum or difference uses their maximum. For this circuit the final bound is

$$456(7964) + 896(4229) + 24(233) + 1760 = 7\,428\,120.$$

An analytic leading-coefficient certificate gives the matching lower bound. Under the diagonal specialization, the source leading coefficients are

$$x_0 = 3^{7964}\beta_S^{3982}, \qquad y_0 = 3^{4229}\beta_S^{2129}2^{\kappa-21}, \qquad c_0 = 3^{233}\beta_S^{116}.$$

The scaled radicand A_2 has degree $1\,857\,030$ and leading coefficient $2^{362}3^8x_0^{114}y_0^{224}c_0^6$. It uniquely dominates V_n, while A_2^2 dominates W_n. The final subtracted term has degree $5\,735\,370 < 7\,428\,120$, so A_2^4 is the leading term of F_{10}. Its diagonal leading coefficient is

$$2^{1448}3^{32}x_0^{456}y_0^{896}c_0^{24} = 2^{3727056\kappa-17368}3^{7426392} > 0.$$

Thus no cancellation can lower the certified degree.

Independently, after setting all ten variables equal to one indeterminate t, the verifier propagates the coefficient of the certified upper degree through every gate modulo a prime. It uses the actual fixed coefficients, authenticated against their complete compressed binary binding. The final leading coefficients are

p	101	1009	1000003
$[t^{7428120}]F_{10}(t,\dots,t) \bmod p$	95	646	515853.

These residues also agree with the displayed analytic coefficient. Any one nonzero residue is a second certificate of the matching lower bound. This is an exact leading-coefficient computation, not a sample equality of polynomial values. Separately, exact sparse integer arithmetic verifies the source coefficient identities, all 59 local circuit identities, the Lucas factorization, and the eight-factor norm identity. Those identity checks establish that the circuit computes (18).

The machine-readable circuit and replay files are in the companion `sun/` component. Its evaluator has 337 multiplications, 161 additions, and 27 subtractions. The verifier imports no circuit builder and passes both ordinary Python and `python -O`. Finite borrow, population, and source-assignment tests supplement the unbounded proofs; they do not decide consistency or replace the inherited arithmetic theorems.

7 Further research questions

The present results suggest several concrete next problems.

1. Can a joint search over variable positions, target positions, and compatible shared coefficients improve on the supplied certificates? The search must retain all exact source identities and positivity bounds; height alone is not the objective because final arithmetic cost also depends on exponentiation chains.
2. Can source equations be transformed by small invertible row operations to create more equal kernel coefficients without increasing coefficient heights or breaking the calibrated source interface? The current construction already shows why private positions are stronger than correctness requires.
3. Can the two successive congruence recoveries in the integral carrier be extended to absorb another quotient or another supplied root? The order is essential: first recover the large modulus for the Pell index, then use the exact index to recover scale divisibility.
4. Can one retain the fourteen-variable degree bound while reducing the additional arithmetic gates? Any proposed simplification must address both sign selection and positivity of reconstructed canonical witnesses.
5. Can the new finite source certificate and integral-carrier proof be formalized against the active ProveIt interfaces, then joined to the native compiler and exported as an object-level ZFC proof? The current closed formal consistency equations are useful reference endpoints, but cannot be substituted for this task.
6. Can a substantially smaller direct ZFC proof verifier change the source polynomial enough to beat these constructions without the K215 route? Merely replacing a literal machine by an unspecified universal polynomial does not provide a comparable explicit equation.

These are research questions, not claimed consequences of the present certificates. In particular, no eight-natural-variable or nine-integer- variable result, no universal optimality theorem, and no consistency decision follows from the improvements reported here.

8 Reproduction and integration

The archive contains this paper's complete LaTeX source, its compiled PDF, the three mathematical components, exact fixed-coefficient recipes and payloads, source provenance, and an integrated verification entry point. The top-level `README.md` gives the tested commands and identifies the primary circuits. Run the supplied verification entry point from the extracted archive; the finite checks do not require executing a ZFC refutation search. Rebuilding the paper uses two passes of `pdflatex -halt-on-error report.tex`.

The repository comparison was frozen before the new work. The complete snapshot identifier is given in the abstract and in the provenance file. No remote repository edits are part of this delivery. The components keep their original source licenses and notices. Newly written material is provided under the repository-compatible license included in the archive.

References

[1] V. Reshetnikov and contributors, *ProveIt: DiophantineConsistency*, commit `beb0192cb8dc36f78c369c080f2adfbc1887a486`. https://github.com/VladimirReshetnikov/ProveIt/tree/beb0192cb8dc36f78c369c080f2adfbc1887a486/Computability/DiophantineConsistency.

[2] ProveIt, source 31, *Con(ZFC) in Diophantine form*, source-31 quadratic system and native coefficient binding, at the snapshot in [1].

[3] ProveIt, source 44, *Sparse B4 encodings*, compact native coefficient tuple and compiler proof, at the snapshot in [1].

[4] ProveIt, source 51, *Sparse arithmetic*, squared-modulus auxiliary quotient and 66-operation construction, at the snapshot in [1].

[5] ProveIt, source 54, *Calibrated carry encodings*, ten-integer arithmetic-cost comparison, at the snapshot in [1].

[6] ProveIt, source 56, *Free Pell coordinates*, free-coordinate and carry construction, at the snapshot in [1].

[7] ProveIt, source 57, *Separated degrees and Pell quotients*, at the snapshot in [1].

[8] ProveIt, source 58, *Integral-root carriers for Con(ZFC)*, at the snapshot in [1].

[9] ProveIt, source 59, *Degree-separated encodings and an ordinary-root merger for Con(ZFC)*, at the snapshot in [1].

[10] J. P. Jones, Universal Diophantine equation, *Journal of Symbolic Logic* **47** (1982), 549–571.

[11] J. Bayer, M. David, M. Hassler, Yu. Matiyasevich, and D. Schleicher, *Diophantine equations over $\mathbb{Z}$: universal bounds and parallel formalization*, arXiv:2506.20909v2 (2025). https://arxiv.org/abs/2506.20909v2.

[12] Z.-W. Sun, *Ten unknowns for Hilbert's tenth problem over the integers*, arXiv:2609.01594v2 (2026). https://arxiv.org/abs/2609.01594v2.

[13] Z.-W. Sun, Further results on Hilbert's tenth problem, *Science China Mathematics* **64** (2021), 281–306. https://arxiv.org/abs/1704.03504v7.